

EveNet for κ_λ Density-Ratio Learning in $HH \rightarrow b\bar{b}\tau^+\tau^-$: Pre-training Improves Discrimination but Degrades Low-Statistics Ratio Calibration

Working note — $b\bar{b}\tau^+\tau^-$ κ_λ NSBI project

16 July 2026 (rev. 14 August: (P5), Sec. 7; 20 August 2026: verdict, Sec. 8)

Gates G0 / G0.5 of `EVENET_FEASIBILITY_NOTE`; 90-run sweep, branch `fm`

Abstract

We test whether the pre-trained event-level foundation model (FM) EveNet [1] improves the *data efficiency* of density-ratio learning for a Higgs self-coupling (κ_λ) measurement by neural simulation-based inference (NSBI), on CMS full-simulation $HH \rightarrow b\bar{b}\tau^+\tau^-$ events. Three arms — fine-tuned, frozen-backbone and from-scratch — identical in data, architecture and head and differing *only* in initialisation, were trained at six training fractions $\times$ five seeds (90 runs) and evaluated on one fixed held-out split against two axes fixed before unblinding: discrimination (weighted area under the receiver-operating characteristic curve, AUC) and ratio calibration (integral and shape closure). Three results. **(i) Discrimination: pre-training wins decisively** — +0.014 to +0.017 AUC at every fraction $\geq 1\%$, worth $\approx 3\text{--}4.5\times$ the training data, with both pre-trained arms exceeding a feature-level gradient-boosted decision tree (BDT) ceiling. **(ii) Calibration quality: pre-training wins at full statistics** — with five-model ensembling the frozen-backbone arm is the *only* configuration in the study that passes both analysis gates, and both pre-trained arms show $\approx 3\text{--}4\sigma$ less residual structure than from-scratch. **(iii) Calibration data efficiency: resolved — no improvement, and measurable harm for the frozen arm** — the (P5) $K=20$ campaign (225 further trainings, paired test-set bootstrap; Sec. 8) gives $\Delta_{\text{finetune}} = -0.022 \pm 0.026$ (bounded: $|\Delta| \lesssim 0.05$) and $\Delta_{\text{frozen}} = -0.079 \pm 0.031$ ($z = -2.6$, crossing the pre-registered harm threshold), with the secondary shape statistic agreeing ($z = -2.0$ and -3.2). On the same ensembles the AUC advantage persists, so pre-training buys event ordering while *costing* low- N calibration — consistent with an initialisation bias in the weakly-pinned normalisation direction (Sec. 7) that ensembling cannot remove and that full statistics eventually repins: at fraction 1.0 the frozen ensemble remains the study’s only gate pass. In both (i) and (ii) the frozen arm matches or beats fine-tuning, locating the benefit in the pre-trained *representation* rather than in backbone adaptation. Section 9 documents three conversion errors found and fixed during analysis, one of which had inflated all closure numbers by a factor of ≈ 3 ; a fourth hypothesis, that EveNet’s diffusion-noised objective causes miscalibration, was refuted by source audit before the ablation was believed.

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1 Setup

Task. Learn per-event density ratios $r(x; \kappa_\lambda, \kappa'_\lambda)$ for an unbinned NSBI likelihood. The κ_λ dependence is a *continuous* deformation of one process’s spectrum (the m_{HH} lineshape and triangle–box interference), not the separation of two distinct processes.

Data. CMS 2024 full-simulation $HH \rightarrow b\bar{b}\tau^+\tau^-$ Powheg samples through the KIT CROWN chain (published $\mu\tau_h/e\tau_h$ selection), converted to EveNet’s stock 7-feature token schema: four tokens per event (b_1, b_2, τ_h, ℓ), missing transverse energy in the global conditioning block, weights = **genWeight** $\times$ **puweight** (signed; the reference carries 6.1% negative weights, the hypothesis 1.0%). Requiring a valid $H \rightarrow b\bar{b}$ candidate drops 18.6–24.5% of events (κ_λ -dependent). Class 0 = $\kappa_\lambda = 1$ (reference, 488k), class 1 = $\kappa_\lambda = 5$ (339k); merged, shuffled, split 80/10/10 into 661.9k train / 82.7k validation / 82.7k test. Training fractions multiply the *train* split; **all** runs share the one fixed test split.

Arms. **finetune** (20M pre-trained checkpoint, all weights trainable); **frozen** (same checkpoint, backbone GlobalEmbedding/PET/ObjectEncoder frozen, heads trainable — isolates the representation from optimiser effects); **scratch** (random init). Fractions $\{0.003, 0.01, 0.02, 0.05, 0.1, 1.0\}$ ($\approx 2.0\text{k}$ to 662k training events) $\times 5$ seeds = 90 runs. Options files are the authors’ own downstream recipes, vendored unchanged apart from the freeze flags.

External bar. A gradient-boosted decision tree on the 10 high-level features behind the published unbinned result (Sec. 6).

2 Pre-registration

Fixed before any sweep output existed, reproduced unchanged:

- (P1) **Decision primacy.** The adoption metric is the *closure left-shift*: the smallest fraction at which an arm passes both gates, compared between finetune and scratch; their ratio is the

equivalent-data multiplier. A left-shift is an adoption-grade win *even at AUC parity*; an AUC-only win with failing closure is *not* a win. AUC is a secondary no-regression guard.

- (P2) **Gates.** $|\text{IC}| \leq \max(1\%, 3\sigma_{\text{stat}})$ and $\text{SC}_{\text{rms}} \leq 0.05$ (= the analysis CLOSURE_TOL); $\chi^2/\text{ndf} \approx 1$ indicates noise-only residuals.
- (P3) **Checkpoint selection.** Every arm is evaluated at its best-validation-loss checkpoint (EveNet’s predictor otherwise loads the final, most overfit, epoch).
- (P4) **Fallback.** If no arm passes anywhere, leaving the left-shift undefined, the secondary reading is relative closure at matched fractions, reported as such.
- (P5) **Seed-extension decision rule** (*added 14 August 2026; fixed before any $K > 5$ output existed*). The sweep is extended by seeds 5–19 at fractions $\{0.003, 0.01, 0.02, 0.05, 0.1\}$ — 225 runs, identical data, recipe, best-validation selection and test split — giving $K=20$ ensembles per cell there. For each pre-trained arm $a \in \{\text{finetune}, \text{frozen}\}$ define

$$\Delta_a = \frac{1}{5} \sum_f (|\text{IC}|_f^{\text{scratch}} - |\text{IC}|_f^a), \quad \sigma_\Delta^2 = \frac{1}{25} \sum_f \left[\sigma_{\text{boot},f}^2 + \frac{\hat{\sigma}_{a,f}^2 + \hat{\sigma}_{\text{scratch},f}^2}{20} \right],$$

where $|\text{IC}|_f$ are the $K=20$ ensemble values, $\sigma_{\text{boot},f}$ is the standard deviation over 200 Poisson($\mu=1$) test-set replicas of the *paired* arm difference (common draws across arms at each fraction), and $\hat{\sigma}$ are the per-seed spreads. Verdicts: $\Delta_a > 2\sigma_\Delta$ — pre-training improves ratio calibration; $\Delta_a < -2\sigma_\Delta$ — it harms it; otherwise any effect is bounded and quoted as $|\Delta_a| \lesssim 2\sigma_\Delta$ (forecast ≈ 0.05). The same statistic on SC_{rms} is reported as secondary, not decisive. All three outcomes are reportable; none will be reinterpreted after unblinding. [*Executed 20 August 2026; verdict in Sec. 8.*]

(P4) is the branch that materialised for the finetune–scratch comparison. (P5) was registered after the $K=5$ results below were known, but before any extension job was submitted; the $K=5$ data motivated the campaign and play no further role in the verdict.

3 Metrics

3.1 The closure identity

The ratio estimate is $\hat{r} = \frac{p}{1-p} \cdot f_{\text{prior}}$, with f_{prior} fixed by the convention EveNet’s loss implies (Sec. 9). Everything rests on one identity: averaging the *true* ratio over reference events,

$$\mathbb{E}_{\text{ref}}[r] = \int p_{\text{ref}}(x) \frac{p_{\text{hyp}}(x)}{p_{\text{ref}}(x)} dx = \int p_{\text{hyp}}(x) dx = 1. \quad (1)$$

The reference density cancels and what remains integrates to one. Intuitively $r(x)$ says “this event is r times more likely under the hypothesis”; over the reference ensemble the $r > 1$ and $r < 1$ regions must balance exactly. On weighted Monte Carlo the expectation becomes the self-normalised weighted average $\sum_{i \in \text{ref}} w_i \hat{r}_i / \sum_{i \in \text{ref}} w_i$, converging at $1/\sqrt{n_{\text{eff}}}$ — so closure is always “consistent with 1”, never “equal to 1”.

3.2 The two metrics

Integral closure, $IC = \mathbb{E}_{\text{ref}}[\hat{r}] - 1$

Reweight the reference by $\hat{r}$ and ask whether the *total* hypothesis yield is recovered. $IC = +0.20$ means a 20% over-production. This is the *normalisation* error. Quoted with its weighted standard error.

Shape closure, SC_{rms} and SC_{norm}

IC can be satisfied by errors that cancel globally, so Eq. (1) is applied *differentially*: quantile-bin the score, and in each bin compare the $\hat{r}$ -reweighted reference mass A_b with the hypothesis mass B_b , forming $\text{rel}_b = A_b/B_b - 1$. We report the RMS over bins with each bin’s statistical variance subtracted in quadrature. We bin in the classifier score because it is monotone in $\hat{r}$, so miscalibration is forced to appear there.

Why the per-bin error is $1/n_{\text{eff}}$. For a weighted sum $S = \sum_i v_i$ with independent (Poisson) presence, $\text{Var}(S) = \sum_i v_i^2$, so

$$\frac{\text{Var}(S)}{S^2} = \frac{\sum_i v_i^2}{(\sum_i v_i)^2} \equiv \frac{1}{n_{\text{eff}}}, \quad n_{\text{eff}} \equiv \frac{(\sum_i v_i)^2}{\sum_i v_i^2}, \quad (2)$$

which *defines* the effective sample size (Kish). Unweighted events give $n_{\text{eff}} = N$ and the Poisson $1/\sqrt{N}$; unequal weights always reduce it ($v = (10, 1, 1, 1)$ gives $n_{\text{eff}} = 1.6$, four events behaving like 1.6). On the reference side $v_i = w_i \hat{r}_i$, so the reweighting inflates the spread — which is why bins at $\hat{r} \rightarrow \infty$ are dropped by the n_{eff} threshold rather than allowed to dominate. Since rel_b is a ratio of estimates from two *independent* samples, $\sigma_b^2 = 1/n_{\text{eff},0} + 1/n_{\text{eff},1}$.

Why subtract in quadrature. With $\text{rel}_b = \beta_b + \varepsilon_b$ (β bias, ε noise of variance σ_b^2), $\mathbb{E}[\text{rel}_b^2] = \beta_b^2 + \sigma_b^2$: the naive RMS is inflated by noise, worst at small statistics. Subtracting the known σ_b^2 estimates the bias RMS, $SC_{\text{rms}} = \sqrt{\max(\langle \text{rel}^2 \rangle - \sigma_b^2, 0)}$. The clipping at zero is why exact 0.000 entries appear: they mean “no bias resolvable above the statistical floor”. Without the subtraction the metric rejects correct models — the first implementation did exactly that, failing an analytically calibrated toy in three seeds of four. The same σ_b normalises $\chi^2/\text{ndf} = \langle (\text{rel}_b/\sigma_b)^2 \rangle$.

3.3 SC_{rms} versus SC_{norm}

SC_{rms} is computed from $\hat{r}$ as delivered, SC_{norm} from $\hat{r}/\mathbb{E}_{\text{ref}}[\hat{r}]$. For a *pure* scale error $\hat{r} = c r_{\text{true}}$, every bin has $A_b/B_b = c$, so $SC_{\text{rms}} \simeq |c - 1| \simeq |IC|$ although the shape is perfect, while $SC_{\text{norm}} = 0$. The two raw numbers are therefore not independent, and reading them together gives three signatures:

signature	IC	SC_{rms}	SC_{norm}	diagnosis
calibrated	≈ 0	≈ 0	≈ 0	usable
pure scale error	$ c - 1 $	$\approx IC $	≈ 0	normalisation only — removable
shape distortion	any	large	large	deformed likelihood — not removable

Table 1: Validated on synthetic data: a $\times 1.4$ scale error gives (0.399, 0.399, 0.000); a temperature-1.5 distortion gives (0.450, 0.694, 0.503).

Why the distinction is physical. Process p enters the likelihood as $\nu_p p_p(x)$ with ν_p the yield (from cross section $\times$ luminosity $\times$ acceptance, determined independently of the classifier) and $p_p = r_p p_{\text{ref}}$. If $\hat{r}_p = c r_p$ then $\nu_p \hat{p}_p = (c\nu_p)[\hat{p}_p/c]$: the constant is algebraically indistinguishable from mis-stating the *yield*, leaving the normalised shape untouched. It is therefore removable by construction (rescale by $\mathbb{E}_{\text{ref}}[\hat{r}]$ on held-out events) and, uncorrected, propagates only through that process’s normalisation where it is degenerate with a rate nuisance parameter. For a coupling measurement this is not harmless — $\sigma(\kappa_\lambda)$ carries the physics — but it is a one-parameter defect with a one-parameter cure. A shape error is different in kind: $\hat{r}(x)/r(x)$ varies with x , events are mis-weighted relative to one another, and no constant or rate parameter repairs it. SC_{rms} is the pre-registered gate; SC_{norm} is a post-hoc decomposition reported alongside.

What closure cannot catch. Both metrics test *self-consistency*, not *sufficiency*. A classifier on lossy features $\phi(x)$ converges to the exact ratio of the *marginalised* densities, satisfies Eq. (1) identically, and passes any test binned in a function of ϕ — the degenerate $\hat{r} \equiv 1$ passes both metrics perfectly while carrying no information. Information loss appears not as bias but as a *flatter* ratio, i.e. lost discrimination. Closure catches lies, not omissions: it earns $\hat{r}$ the right to enter the likelihood, while AUC shows it is worth entering.

4 Result 1 — discrimination (G0)

fraction	single model (seed mean $\pm$ s.d.)			5-seed ensemble		
	finetune	frozen	scratch	finetune	frozen	scratch
0.003	0.7427(57)	0.7498(80)	0.7573(122)	0.7553	0.7674	0.7698
0.01	0.7768(108)	0.7793(76)	0.7612(300)	0.7924	0.7914	0.7787
0.02	0.7941(43)	0.7931(80)	0.7804(19)	0.8040	0.8041	0.7845
0.05	0.8056(54)	0.8099(22)	0.7949(8)	0.8144	0.8160	0.7983
0.1	0.8163(22)	0.8169(9)	0.8003(25)	0.8210	0.8212	0.8059
1.0	0.8313(2)	0.8313(5)	0.8250(4)	0.8332	0.8330	0.8285

Table 2: Weighted AUC. Parenthesised digits are the seed standard deviation in units of the last decimals.

Figure 1 shows the same at ensemble level, for both campaigns.

1. **Pre-training wins decisively.** At fractions 0.01–1.0 both pre-trained arms exceed scratch by 0.014–0.017 AUC against seed standard deviations of 0.001–0.008 — many standard deviations, monotone across the grid.
2. **Worth $\approx 3\text{--}4.5\times$ the data.** Interpolating scratch’s ensemble curve logarithmically, finetune’s AUC at 0.01, 0.02, 0.05, 0.1 requires scratch fractions of 0.033, 0.079, 0.222, 0.437 — multipliers 3.3, 4.0, 4.4, $4.4\times$; at full statistics scratch never reaches finetune’s value.
3. **Frozen $\approx$ finetune everywhere** (differences ≤ 0.004). The benefit lives in the pre-trained *representation*; adapting the backbone adds nothing measurable. This answers the “did the pretext retain κ_λ -relevant continuous information?” question: it did.

The 0.003 inversion. At the smallest fraction scratch leads. That tier is an optimisation regime, not a representational one: with $\approx 2\text{k}$ events and a global batch of $\approx 8\text{k}$ across 4 GPUs, one “epoch” is

Discrimination: pre-training wins at every fraction ≥ 0.01 (K=20 bars: test-set bootstrap)

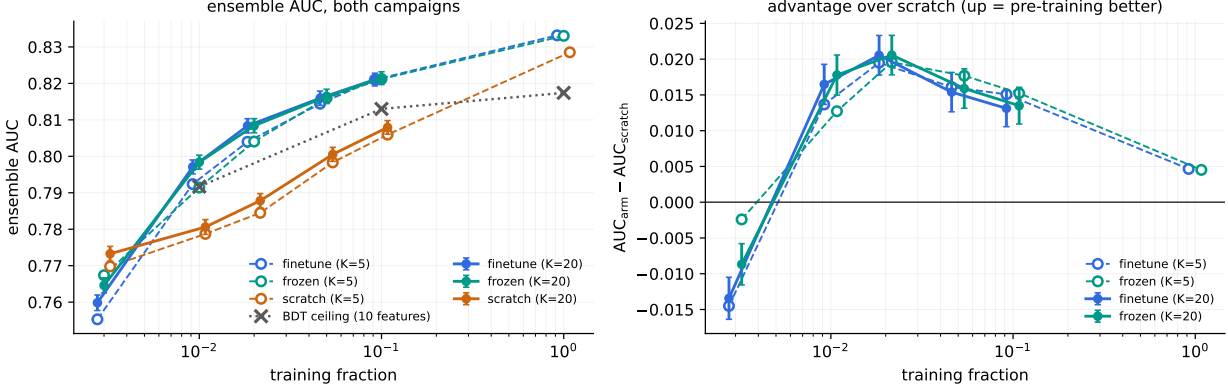


Figure 1: Ensemble AUC versus training fraction for both campaigns (open: 5-seed sweep; filled: 20-seed extension, bars = test-set bootstrap) against the BDT ceiling (left), and the advantage over scratch (right). Pre-training wins at every fraction ≥ 0.01 ; the 0.003 tier is the optimiser-transient inversion discussed below.

a single gradient step, and the selected checkpoints sit at epochs 4–21, i.e. the models are compared after $\mathcal{O}(10)$ updates under *different* learning-rate recipes. We treat it as a recipe artefact pending a batch-size/learning-rate ablation.

5 Result 2 — calibration (G0.5)

All numbers below use the ratio convention derived in Sec. 9. Production NSBI uses ensembles (the analysis config trains five members) and the BDT ceiling is itself an ensemble, so the **ensemble row is the like-for-like object**; single-model numbers are given for reference.

fraction	single model, $ IC $ (seed mean)			single model, SC_{norm}		
	finetune	frozen	scratch	finetune	frozen	scratch
0.003	0.136	0.310	0.164	0.328	0.476	0.365
0.01	0.110	0.279	0.213	0.346	0.341	0.206
0.02	0.148	0.120	0.129	0.276	0.244	0.187
0.05	0.102	0.099	0.158	0.178	0.206	0.159
0.1	0.098	0.143	0.192	0.176	0.168	0.180
1.0	0.050	0.053	0.055	0.090	0.061	0.076

Table 3: Single-model closure. Statistical errors on $|IC|$ are 0.009–0.021. No single model passes the gates at any fraction.

(a) **Ensembling works — once the convention is right.** Because $\mathbb{E}_{\text{ref}}$ is linear in r , $IC_{\text{ens}} = \frac{1}{K} \sum_k IC_k$ identically: the ensemble reduces $|IC|$ only if the members’ offsets differ in sign. Under the superseded convention 15 of 18 cells were same-sign and ensembling did nothing. With the corrected convention the residuals straddle zero (several signed means are negative) and the ensemble collapses them: finetune at full statistics goes $0.050 \rightarrow \mathbf{0.0198}$ (statistical error 0.0186, i.e. consistent with zero), scratch at 0.01 goes $0.213 \rightarrow 0.068$.

arm	fraction	IC	gate	SC _{rms}	gate	verdict
finetune	1.0	0.0198	0.0559 ✓	0.0521	0.05 ×	fails by 4%
frozen	1.0	0.0527	0.0536 ✓	0.0272	0.05 ✓	PASSES
scratch	1.0	0.0362	0.0516 ✓	0.0742	0.05 ×	fails

Table 4: Ensemble closure at full statistics against the pre-registered gates ($|IC| \leq \max(1\%, 3\sigma_{\text{stat}})$, $SC_{\text{rms}} \leq 0.05$). No arm passes at any smaller fraction. χ^2/ndf : finetune 1.80, frozen 1.52, scratch 3.20.

(b) At full statistics the pre-trained arms are better calibrated in shape. Ordering at fraction 1.0 is consistent across all three shape measures: frozen < finetune < scratch on SC_{rms} (0.027/0.052/0.074) and on χ^2/ndf (1.52/1.80/3.20). With 20 bins, $\sigma(\chi^2/\text{ndf}) \approx \sqrt{2/20} = 0.32$, so scratch’s residuals are **3–4 σ** more structured than either pre-trained arm’s. **The frozen ensemble is the only configuration in the study that passes both gates** (Table 4).

(c) But there is no data-efficiency gain — the pre-registered metric is a null. (P1) compares finetune against scratch: neither passes at *any* fraction, so both first-passing fractions are undefined and no equivalent-data multiplier exists. Under the (P4) fallback the reduced-fraction ordering is mixed — scratch is nominally best on SC_{norm} at 0.01–0.05, the pre-trained arms at 0.1–1.0, and frozen is the *worst* arm at 0.003–0.01 while being the best at 1.0. There is no left-shift anywhere. Figure 2 (top row) shows this ordering at $K=5$; its bottom row is the $K=20$ resolution of Sec. 8.

Summary. A usable density ratio requires the full 662k training events *and* ensembling. At that point pre-training measurably improves ratio shape; it does not reduce the data needed to get there — Sec. 8 sharpens this to a signed statement.

6 The ceiling comparison

The feature-level BDT, measured at full statistics with class-normalised weights (so its prior factor is unity by construction):

fraction	AUC	IC	shape
0.01	0.7917	0.202	0.391
0.1	0.8130	0.011	0.039
1.0	0.8174	0.018	0.000

Table 5: Kinematic ceiling (10 published features, gradient-boosted trees).

Discrimination: EveNet wins. At 0.1 and 1.0 both pre-trained arms exceed the ceiling (0.821/0.821 vs. 0.813; 0.833 vs. 0.817), i.e. the object-level model extracts κ_χ information the 10 aggregate features discard. *Caveat:* the ceiling was measured on the un-cut flat-feature population whereas the sweep applies the valid- b -pair requirement, so the sweep population is slightly easier; the arm-to-arm comparisons share identical events and carry no such caveat.

Calibration: comparable, once conventions match. At full statistics the ceiling reaches $|IC| = 0.018$ and shape 0.000; the EveNet ensembles reach $|IC| = 0.020$ – 0.053 and $SC_{\text{rms}} = 0.027$ – 0.074 . The frozen ensemble (0.053, 0.027) is within a factor of a few of the BDT on both axes. An

Ensemble ratio-closure vs training statistics, the two campaigns kept separate

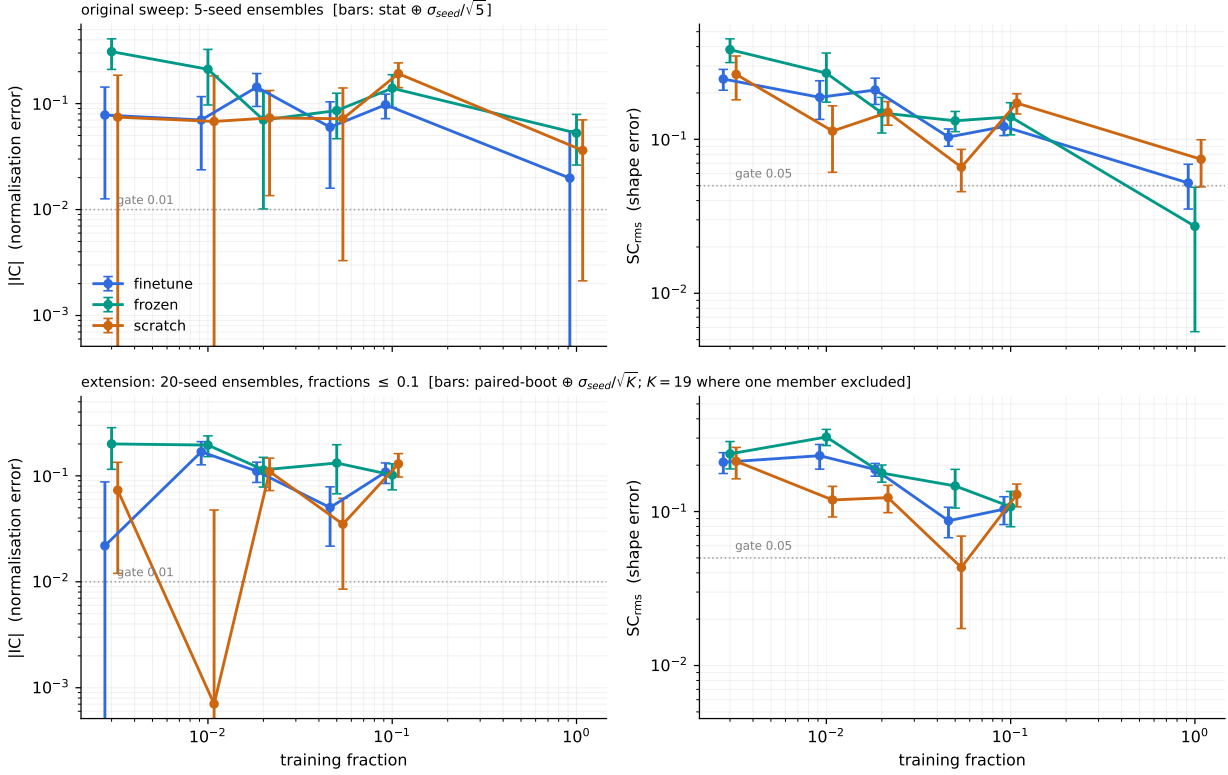


Figure 2: Ensemble $|IC|$ (left) and SC_{rms} (right) versus training fraction. Top: the 5-seed sweep, bars = statistical $\oplus$ seed/ $\sqrt{5}$. Bottom: the (P5) 20-seed extension at fractions ≤ 0.1 , bars = paired test-set bootstrap $\oplus$ seed/ $\sqrt{K}$ (K the cell’s member count; the SC_{rms} seed term is the single-model spread proxy). The campaigns are never joined across the rows.

earlier draft of this note reported the BDT as $\approx 8\times$ better calibrated *despite worse AUC* and drew a strong conclusion from it; that contrast was an artefact of converting the two estimators under different conventions (Sec. 9) and does not survive.

7 Why calibration is harder to power than discrimination

The split verdict of Secs. 4–5 — discrimination resolved decisively by the same 90 runs that leave the calibration ordering ambiguous — is not an accident of this dataset. The two metric families admit structurally different amounts of noise, on both the training and the test side. This section records the mechanisms and the measurements behind that statement: they fix the design of (P5), and they generalise to any foundation-model claim about NSBL.

(a) AUC is a rank statistic; IC is exponentially sensitive to the coordinate AUC discards. AUC is invariant under every monotone remapping of the score: it sees only the ordering of test events. Decompose the difference between two seeds’ learnt logit functions into a monotone part — a global offset δ and gain — and an order-changing part. The monotone part moves IC directly

($r \propto e^f$, so $\text{IC} \approx e^\delta - 1 \approx \delta$) and moves AUC by exactly zero; the order-changing part, which is all AUC can see, is small, because two networks trained on the same data mostly disagree about *how confident* to be, not about which event is more signal-like. Measured on the sweep’s own 90 networks (Table 6), the per-seed spread of IC is 13–160 $\times$ that of AUC, the disparity *growing* with training statistics.

fraction	seed s.d. of IC (ft/fr/scr)	seed s.d. of AUC (ft/fr/scr)	ratio
0.003	0.145 / 0.219 / 0.247	0.006 / 0.008 / 0.012	24 $\times$
0.01	0.098 / 0.252 / 0.256	0.011 / 0.008 / 0.030	13 $\times$
0.02	0.104 / 0.131 / 0.131	0.004 / 0.008 / 0.002	26 $\times$
0.05	0.094 / 0.081 / 0.152	0.005 / 0.002 / 0.001	39 $\times$
0.1	0.040 / 0.098 / 0.105	0.002 / 0.001 / 0.003	43 $\times$
1.0	0.068 / 0.044 / 0.066	0.0002 / 0.0005 / 0.0004	161 $\times$

Table 6: Per-seed spread of the single-model IC versus AUC, per (arm, fraction) cell; the ratio column is the mean-over-arms ratio. Training data grows 330 $\times$ down to the last row: the AUC spread collapses $\approx 30\times$, the IC spread only $\approx 3\times$.

(b) The offset is weakly pinned by the loss, so training noise concentrates in it. Expanding the weighted cross-entropy about its minimum, a global logit shift δ costs $\Delta L \approx \frac{1}{2} \delta^2 \sum_i w_i c_{y_i} p_i (1-p_i)$: the curvature is carried only by events near the decision boundary, and for a classifier at AUC 0.83 most events are confident, $p(1-p) \rightarrow 0$. Numerically, $\delta = 0.05$ (a 5% IC error, the size of the full-statistics seed scatter) costs $\Delta L \sim 10^{-4}$ — invisible to early stopping and to best-validation checkpoint selection, so the frozen-in offset is effectively a per-seed random draw from the training trajectory. Evaluation adds a further monotone jitter: EveNet predicts from an exponential moving average of the *parameters*, while normalisation-layer running buffers are not averaged, an offset-and-gain perturbation by construction. The diagnostic that separates this from sampling noise is the last row of Table 6: 330 $\times$ more training data shrinks the IC seed spread only $\approx 3\times$ (pure sampling would give $\sqrt{330} \approx 18\times$). The residual is optimisation and selection stochasticity, which no single training removes — and which ensembling *does* remove, because the offsets are approximately zero-centred across seeds (Sec. 5).

(c) The test side compounds it. AUC is an average of a bounded comparison over $N_0 N_1 \approx 1.7 \times 10^9$ pairs: test-statistical error ≈ 0.002 . IC is the weighted mean of the heavy-tailed $\hat{r}$ over 4.9×10^4 reference events: floor 0.012–0.019. Per cell, the measured AUC effect (0.015) stands at $\approx 7\sigma$ of its total noise; a plausible calibration effect (0.03) stands at $< 1\sigma$ of its. The same 90 runs power one axis and not the other by construction of the metrics, not by bad luck.

(d) The regime matters: reweighting versus discrimination. When the two samples are nearly identical — OmniFold-style data-versus-simulation reweighting [3, 4], or a nominal-versus-shifted systematic variation — the classifier sits at $p \approx 0.5$ everywhere, exactly where the offset curvature of (b) is maximal: calibration is *cheap* in that regime, and the H1 measurement indeed reports training and initialisation variance negligible against data statistics. Well-separated signal-versus-background, as here, is the opposite corner. The published foundation-model evidence (validation loss, discrimination, distances between normalised unfolded distributions) therefore does not — and structurally cannot — speak to absolute ratio calibration at low N , which is the axis NSBI uniquely needs. The corollary for this programme: the natural near-term calibration win

Where the calibration error bar comes from: training stochasticity (filled) vs test-sample statistics (open)

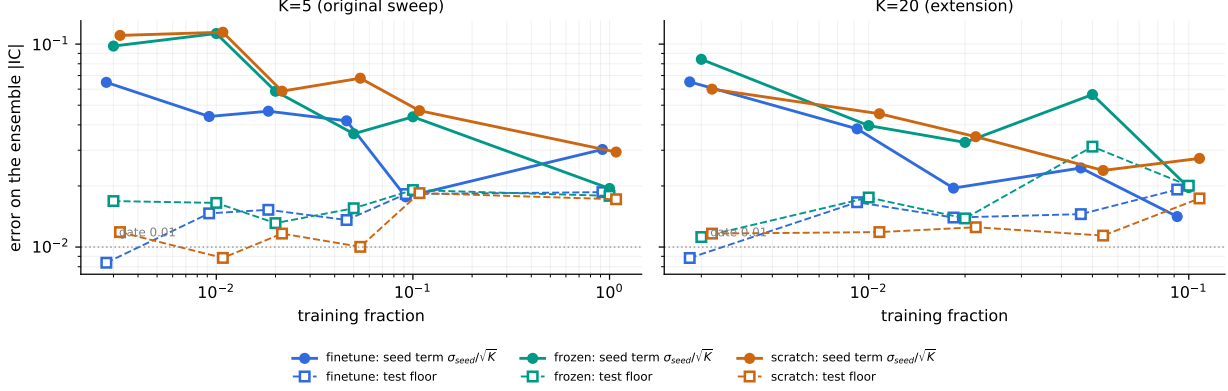


Figure 3: Decomposition of the error on the ensemble $|IC|$ into the training-stochasticity term $\sigma_{\text{seed}}/\sqrt{K}$ (filled) and the test-sample term (open; Gaussian at $K=5$, paired bootstrap at $K=20$). At $K=5$ the seed term dominates by $4\text{--}7\times$ at reduced fractions; at $K=20$ it approaches the test floor, which is where extra seeds stop paying.

for a pre-trained model is the systematic-variation task of N3, whose ratios are near unity, not the signal-extraction ratio.

(e) The arithmetic behind $K=20$. At $K=5$ the seed term exceeds the test floor by $4\text{--}7\times$ at reduced fractions. $K=20$ brings the per-cell error to $0.02\text{--}0.06$, which resolves a consistent arm offset of $\gtrsim 0.05$ at 2σ across the five fractions — the scale suggested by the discrimination gains — while the seed count at which the seed term would meet the test floor is $K^* = (\hat{\sigma}/\sigma_{\text{test}})^2 \approx 40\text{--}400$: seeds beyond ≈ 50 buy little without a larger test split. Pairing the bootstrap draws across arms removes the common test fluctuation from arm differences at zero cost. These three numbers fix the (P5) design (Fig. 3); Sec. 8 reports the executed campaign.

8 Result 3 — the (P5) verdict: no calibration gain at low statistics

Execution. Seeds 5–19 were trained at fractions $\{0.003, 0.01, 0.02, 0.05, 0.1\}$ (225 runs, identical data, recipe, best-validation selection and test split; ≈ 13 GPU-node-hours). One training diverged — finetune, fraction 0.05, seed 12: the validation loss went NaN near epoch 12, *displayed as 0.000*, and early stopping retained the diverged state as the best checkpoint; its predictions were 100% non-finite. Members with non-finite outputs or unreadable predictions are excluded by an objective rule applied identically to all arms; three cells run at $K=19$ (one member lost in each of finetune-0.05, frozen-0.003, scratch-0.1 — one per arm, favouring none). All error propagation uses each cell’s actual K .

Verdict. Applying (P5) exactly as registered (Table 7, Fig. 4): fine-tuning shows *no* calibration data-efficiency gain — any effect is bounded at $|\Delta| \lesssim 0.05$, with a negative central value — and the frozen arm *crosses the pre-registered harm threshold*: its ensemble normalisation error at reduced fractions is worse than from-scratch by 0.079 ± 0.031 , negative in four of the five fractions. The secondary SC_{rms} statistic agrees in sign for both arms and is individually significant for frozen: the pre-trained arms’ ratio *shapes* are also worse than scratch at low N (Table 8). An independent

re-implementation of the verdict arithmetic and a cross-check of every ensemble value against the per-seed closure outputs (linearity, agreement $\sim 10^{-7}$) confirm the numbers.

	$\Delta_{ \text{IC} } \pm \sigma_\Delta$	z	verdict (as registered)	$\Delta_{\text{SC}_{\text{rms}}} \pm \sigma$	z
finetune	-0.022 ± 0.026	-0.9	bounded: $ \Delta \lesssim 0.05$	-0.038 ± 0.019	-2.0
frozen	-0.079 ± 0.031	-2.6	harms calibration	-0.069 ± 0.021	-3.2

Table 7: The (P5) statistics ($\Delta > 0$ would mean the pre-trained arm is better calibrated than scratch). Primary $|\text{IC}|$ column: paired-bootstrap $\oplus$ per-seed terms, as registered. Secondary SC_{rms} : unpaired bootstrap $\oplus$ single-model seed-spread proxy at each cell’s K ; reported, not decisive.

fraction	$ \text{IC} $ (ensemble $\pm$ boot $\oplus$ seed/ $\sqrt{K}$)			SC_{rms}		
	finetune	frozen	scratch	finetune	frozen	scratch
0.003	0.022 ± 0.066	0.200 ± 0.085	0.073 ± 0.061	0.209	0.237	0.212
0.01	0.169 ± 0.042	0.195 ± 0.043	0.001 ± 0.047	0.231	0.305	0.119
0.02	0.111 ± 0.024	0.114 ± 0.036	0.110 ± 0.037	0.188	0.178	0.123
0.05	0.050 ± 0.029	0.132 ± 0.065	0.035 ± 0.027	0.087	0.147	0.043
0.1	0.109 ± 0.024	0.102 ± 0.028	0.130 ± 0.032	0.104	0.107	0.129

Table 8: $K=20$ ensemble closure per cell. The ensemble AUC ordering on the same cells is unchanged from Sec. 4: pre-trained arms $+0.013$ to $+0.021$ above scratch at every fraction ≥ 0.01 (and -0.009 to -0.014 at the optimiser-transient 0.003 tier).

Interpretation. The two axes of Sec. 7 now carry opposite signs on the same networks and the same test events: pre-training improves event *ordering* at every usable fraction while degrading the absolute ratio normalisation and shape at low N . The mechanism follows Sec. 7(b): the calibration offset is weakly pinned by the loss, so the offset the *initialisation* carries survives small- N training — a from-scratch network starts unbiased and settles near its own data’s cross-entropy balance, while the pre-trained backbone imports a normalisation prior from a different corpus that 2k–66k events cannot repin. Crucially this is an arm-level *bias*, not zero-centred seed noise: ensembling — which resolved everything at $K=5$ — cannot remove it, and it became visible precisely when $K=20$ beat the seed noise below its size. It is a low- N phenomenon: at full statistics the ordering inverts and frozen is the best-calibrated configuration (Sec. 5).

Post-hoc annotations (labelled as such; the verdict is mechanical): (a) the finetune average is dominated by the 0.01 cell and the 0.1 cell is positive — the finetune deficit is fraction-heterogeneous, the frozen deficit is not (negative in 4/5); (b) the scratch 0.01 value $|\text{IC}| = 0.0007$ is a folded $|\cdot|$ statistic sitting on zero with a seed error of 0.045 — its point value should not be over-read, and the (P5) error propagation carries its full uncertainty; (c) at fraction 0.1 both differences turn mildly positive, consistent with the full-statistics inversion.

9 Corrections applied during analysis

Four hypotheses about the apparent calibration failure were tested. Three were our own conversion errors; the fourth was refuted before it was believed. Recorded because the first three each changed the numbers materially and the fourth would have cost 15 GPU jobs.

Same 20-seed ensembles, same test events: opposite verdicts on the two axes (up = pre-training better; bars: paired-bootstrap $\oplus$ seed/ $\sqrt{K}$)

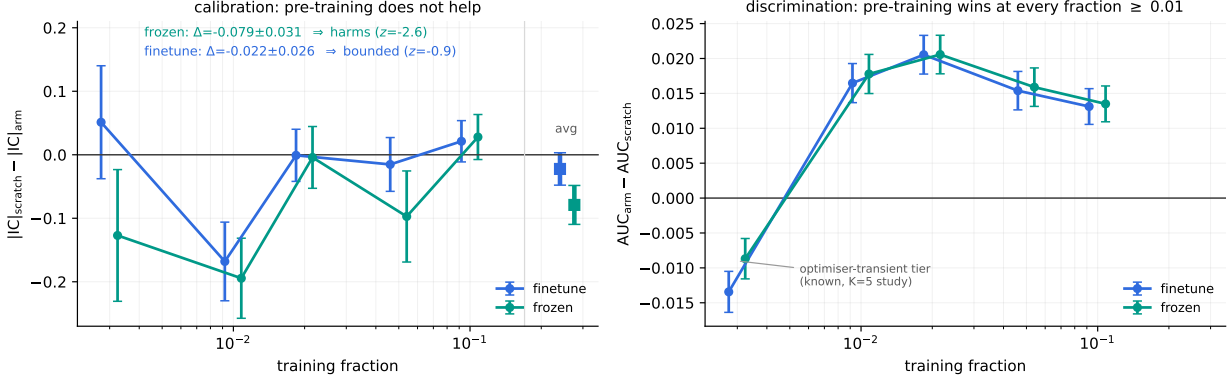


Figure 4: The (P5) verdict on the $K=20$ ensembles. Left: the per-fraction statistic $|IC|_{\text{scratch}} - |IC|_{\text{arm}}$ (up = pre-training better calibrated) with paired-bootstrap $\oplus$ seed errors, and the fraction-averaged $\Delta_a \pm \sigma_\Delta$ as the detached squares. Right: the AUC advantage of the same ensembles on the same test events. One object, two axes, opposite signs.

C1–C3: the ratio prior convention (a chain of three). EveNet’s classification loss is $\sum_i c_{y_i} w_i \text{CE}_i / \sum_i c_{y_i} w_i$, with $c = \text{normalization.pt}["\text{class_balance}"]$ and w the signed per-event weight (`apply_event_weight` is true). Its population optimum is $p/(1-p) = c_1 dW_1(x)/(c_0 dW_0(x))$, and since $r = [dW_1/W_1]/[dW_0/W_0]$,

$$r(x) = \frac{p}{1-p} \cdot \frac{c_0 W_0}{c_1 W_1}, \quad W_c = \text{class_counts}[c] \text{ (signed, train split)}. \quad (3)$$

We used, in order: W_0/W_1 from the test split (C1, wrong — omits c entirely); 1.0 (C2, “class-balanced” — correct only if c were inverse-frequency); and finally Eq. (3) (C3). The key fact is that `class_balance` is *inverse effective number of samples*, not inverse frequency — verified by reconstructing the stored vector bit-for-bit in `float32` from `class_counts` via $\beta = 1 - 1/N$ effective-number reweighting — so it cancels only 77.9% of the class imbalance in log space. Numerically $f = (1.2616 \times 11351.2)/(0.7384 \times 22569.1) = \mathbf{0.8593}$, against an empirically required 0.8702 ± 0.0498 measured over the 15 full-statistics runs (0.22σ ; no arm dependence). C2 inflated $|IC|$ by $\approx 3\times$ and, because a constant scale error shifts every bin equally, inflated SC_{rms} by the same amount — which is why the first pass showed closure *flat* in N rather than converging.

C4: refuted — the diffusion-noised objective. We hypothesised that EveNet’s classification head, trained with `noise_prob=1.0` but evaluated at the clean $t=0$ endpoint, is not a proper scoring rule there, and planned a 15-job ablation with `noise_prob=[0,0]`. Source audit refuted this before the result was believed: `noise_prob` is read at exactly one line, inside a `generation` branch that three flags in our own configs disable (`ReconGeneration.include: false, generation-recon: [0,0], classification-noised: [0,0]`). The branch that ran, `deterministic`, sets `full_time=0` — *identical* to predict time. There was no train/predict mismatch and the ablation would have been a no-op.

Other fixes. Closure is evaluated with signed weights (the measure the loss used); predict configs inherited an unresolvable relative `options.yaml` path and lacked the head-inclusion block, so every

prediction task initially died before loading a checkpoint.

10 Caveats

- **No sensitivity claim.** No σ_{κ_λ} statement is made or implied; the stock schema discards τ -identification and FastMTT $m_{\tau\tau}$ (equally in all arms), and the MC-statistics estimator question is open upstream [5].
- **One channel, one hypothesis pair** ($\kappa_\lambda = 1$ vs. 5; μ_{τ_h}, e_{τ_h}). The $\kappa_\lambda = 0$ pair and the systematic-variation task are untouched.
- **Fraction-1.0 cells remain $K=5$** (the extension covered reduced fractions only); full-statistics comparisons carry the original seed-noise floor.
- **The (P5) verdict is recipe- and task-conditional:** it is measured for this fine-tuning recipe, this hypothesis pair, and this test-set size; a calibration-aware fine-tuning objective (e.g. re-pinning the output layer’s normalisation before training) could change it — that is a constructive question for the authors, not an assumption.
- **Frozen passes by a thin margin** ($|\text{IC}| = 0.0527$ against a gate of 0.0536). It should not be over-read as a robust pass.
- **Population mismatch vs. the ceiling only** (Sec. 6); all arm-to-arm comparisons share identical events.
- **The 0.003 tier** is optimiser-transient-dominated (Sec. 4).

11 Next steps

- N1. [P1, S — done] Bootstrap the ensemble closure metrics: implemented as a $\text{Poisson}(\mu=1)$ test-set bootstrap with draws paired across arms (`eval_ensemble.py --bootstrap`, run automatically by the `eval` stage); the seed-extension campaign it feeds is pre-registered as (P5).
- N2. [P1, S] G1 corner-encoding A/B (τ_h at the vacant flag corner $(0, 0, \pm 1)$): the NPZs already exist for both encodings.
- N3. [P1, M] Money Plot 2 — systematic-variation data efficiency, using the analytic injector where closure is exact by construction. This is the regime the FM’s low-statistics claim was actually aimed at, and it is untested.
- N4. [P2, S] Low-fraction recipe ablation (batch size / learning rate at 0.003) to convert the AUC inversion from artefact to measurement.
- N5. [P2, S] Re-measure the ceiling on the valid- b -pair population to remove the asterisk in Sec. 6.

For the EveNet authors. (i) The ratio conversion of Eq. (3) is not documented anywhere in the repository and is easy to get wrong in three distinct ways (Sec. 9); for any likelihood-ratio use of the classifier head this is the single most valuable thing to document, or to expose as a utility. (ii) Our G1 result will be the first data on reusing *vacant flag corners* for object types absent from pre-training (τ_h, γ) — would they endorse this as the zero-surgery interim? (iii) Would they

consider retrofitting an explicit type-embedding table, initialised from the current flag patterns so it is function-preserving on day one, making future object types an append-a-row operation rather than schema surgery? (iv) Bug report from the (P5) campaign: a diverged training’s NaN validation loss displays as 0.000, which early stopping and best-checkpoint selection then treat as the best score — the diverged state wins checkpoint selection silently. A NaN guard in the trainer (terminate, and never checkpoint a non-finite loss) would fix it. (v) The Sec. 8 result suggests any FM claim aimed at likelihood-ratio use should be powered and reported on the calibration axis, not only on discrimination metrics — and that the near-unity systematic-variation regime, not signal extraction, is where pre-trained calibration gains are most plausible.

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